

RTL to Gate-Level Simulation

Goal

Synthesize RTL to a gate-level netlist using Yosys and verify equivalence through gate-level simulation.

Overview

This project implements a simple OR gate in Verilog, synthesizes it to a gate-level netlist, and compares RTL and gate-level simulation outputs.

RTL Code

```
module or_gate (
    input a,
    input b,
    output y
);
    assign y = a | b;
endmodule
```

Testbench Code

```
module tb_or_gate;
    reg a,b;
    wire y;

    or_gate dut(
        .a(a),
        .b(b),
        .y(y)
    );

    initial begin
        $dumpfile("dump.vcd");
        $dumpvars(0,tb_or_gate);
        $display("Time\tA\tB\tY");

        a=0; b=0; #10;
        $display("%0t\t%b\t%b\t%b", $time, a, b, y);

        a=0; b=1; #10;
        $display("%0t\t%b\t%b\t%b", $time, a, b, y);

        a=1; b=0; #10;
        $display("%0t\t%b\t%b\t%b", $time, a, b, y);

        a=1; b=1; #10;
        $display("%0t\t%b\t%b\t%b", $time, a, b, y);

        $finish;
    end
endmodule
```

Gate-Level Netlist

```
module or_gate(a,b,y);
    input a;
    input b;
    output y;

    or(y,a,b);

endmodule
```

Yosys Synthesis Script

```
read_verilog or_gate.v
synth -top or_gate
write_verilog or_gate_netlist.
```

Expected Simulation Output

Time	A	B	Y
10	0	0	0
20	0	1	1
30	1	0	1
40	1	1	1

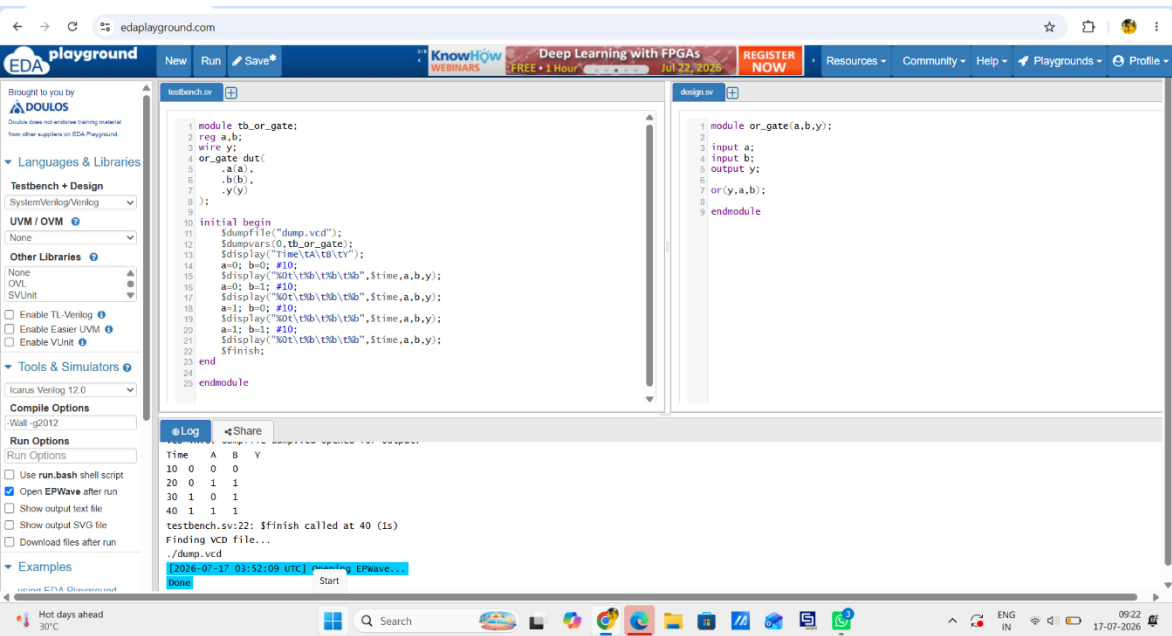
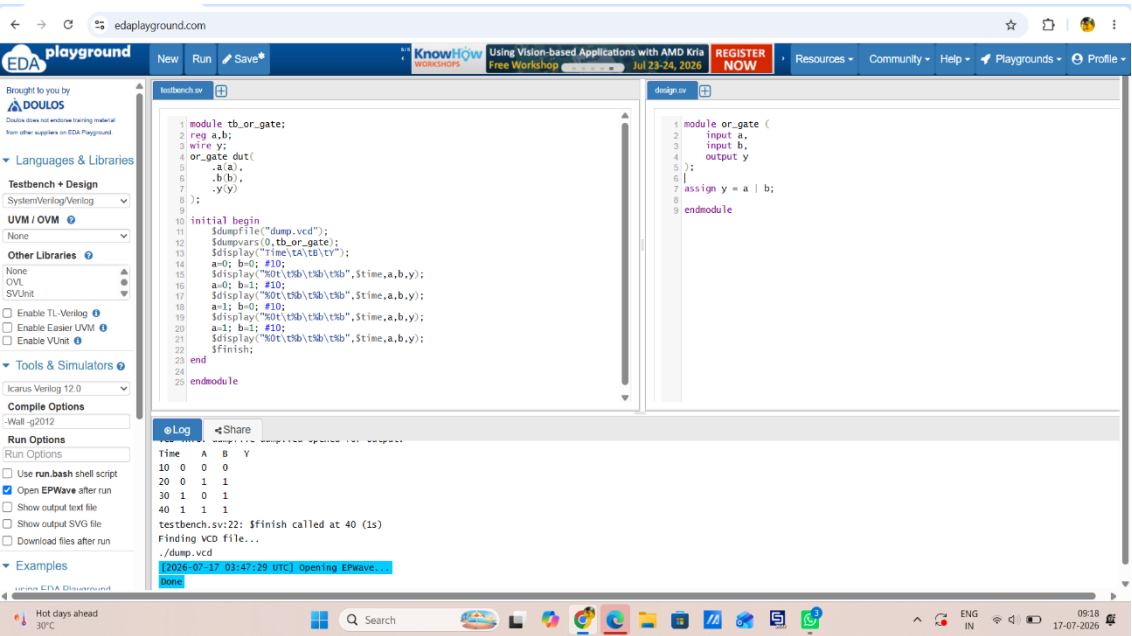
Coverage Explanation

All four input combinations (00, 01, 10, 11) are tested. This provides 100% functional coverage for a 2-input OR gate.

RTL vs Gate-Level Comparison

The RTL implementation uses the Verilog OR operator. The synthesized gate-level netlist uses a primitive OR gate. Both produce identical outputs for all input combinations, demonstrating functional equivalence.

Simulation



Waveforms

